

AXEL GASTALDI

Resume

CONTACT

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RESEARCH DOMAIN

My research lies investigates *homological* and *K-theoretical invariants* of crossed product algebras, bridging **Non-Commutative Geometry** and **Representation Theory**, with a particular focus on *cyclic homology*, *equivariant K-theory*, and the *Connes–Kasparov framework* for reductive Lie groups. I'm currently studying periodic cyclic homology of smooth crossed product algebras associated to Lie groups.

EDUCATION

- 2026/2028: **Post-doctorant** supervised by Pr. Hang Wang at the Research Center for Operator algebras in Shanghai;
- 2022/2026: **PhD in Mathematics** supervised by Pr. Michael Puschnigg at the Institut de Mathématique de Marseille ([PhD Thesis](#));
- 2020/2022: *Master degree* in Fundamental Mathematics *supported by a Master thesis funding* of the Archimede Institute - Aix-Marseille Université
- 2017/2020: *Bachelor degree* in Mathematics - Aix-Marseille Université

PUBLICATIONS

(In redaction, expected in 2026) "*Dirac-dual Dirac method in periodic cyclic homology.*"

2026: "*Periodic cyclic homology of crossed product algebras*" - [PhD Thesis](#).

TALKS

- June 2026: "*An homological analogue of the Baum-Connes conjecture with coefficients for Lie groups*" for the team seminar; Institut de Mathématiques de Marseille
- May 2026: "*Periodic cyclic homology of smooth crossed product algebras*" for the [Spring Program 2026](#); East China Normal University ([Notes](#))
- December 2025: "*Banach Dirac-dual Dirac method in periodic cyclic homology*"; Albi ([Notes](#))
- June 2025: Mini-course about "*Introduction to cyclic homology*"; Université Paul Sabatier ([Notes](#))
- February 2025: "*Mackey analogy in periodic cyclic homology*" for the [Tempered Representation and K-theory](#) thematic semester; Institut Henri Poincaré ([Notes/Video](#))
- December 2024: "*Periodic cyclic homology of crossed product algebras*"; Institut Denis Poisson
- November 2024: Mini course about "*Differentiation and (co)homology*"; Centre Physique Théorique ([Notes/Videos 1 and 2](#))
- February 2023: "*Derived categories and derived cohomology*" for Condensed Maths working groups - Institut de Mathématiques de Marseille

OTHER ATTENDED CONFERENCES

- January 2025: Conference "*Representation Theory and Operator Algebras*"; Centre International de Rencontres Mathématiques
- December 2023: Conference "*Arbre de Noël de Géométrie Non-Commutative*"; Institut Élie Cartan de Lorraine
- March 2023: Conference "*Condensed Maths*"; CIRM
- June 2022: Conference "*Arbre de Noël de Géométrie Non-Commutative*"; Université de Caen

TEACHING EXPERIENCE

Teaching at university:

- 2025/2026:
 - 120h: "General Mathematics": first-year students in Mathematics
 - 3×12h: "Stage Hippocampe": middle-school and high-school students
 - 24h: "Mathematic tools"; first-year students in Biology
- 2020/2021: 24h: "Tutoring in Mathematics"; second-year students in Mathematics

Presentations for PhD seminars:

- May 2026: "*How to study a manifold ?*"; Institut de Mathématiques de Marseille
- October 2025: "*The beautiful Connes-Kasparov Isomorphism*"; Institut de Mathématiques de Marseille
- June 2025: "*Curvatures as topological invariant, the Chern-Weil construction*"; Université Paul Sabatier
- March 2025: "*Gentle introduction to K-theory*"; Institut de Mathématiques de Marseille
- April 2024: "*A story of characteristic classes*"; Institut de Mathématiques de Marseille
- April 2024: "*Group cohomology*"; Institut de Mathématiques de Marseille
- May 2023: "*Algebraic study of topological groups*"; Institut de Mathématiques de Marseille
- March 2023: "*Non-commutative differential geometry in terms of stars*"; Institut de Mathématiques de Marseille

PARTICIPATION TO DOCTORAL COURSES AND WORKING GROUPS

- September - December 2024: "*Classes caractéristiques*"; doctoral course given by Pr. Michael Puschnigg - Institut de Mathématiques de Marseille
- September - December 2024: "*Géométrie Non-Commutative pour la Physique*"; Centre de Physique Théorique
- April 2024: "*Cohomologie galoisienne*"; Institut de Mathématiques de Marseille
- September - November 2023: "*Théorie de Deligne-Lusztig*"; doctoral course given by Pr. Olivier Dudas - Institut de Mathématiques de Marseille
- January 2023 - March 2024: "*Mathématiques Condensées*"; Institut de Mathématiques de Marseille

OUTREACH AND LEADERSHIP

- *Responsible from June 2023 to February 2025* of the [PhD seminar](#) of Mathematics/Physics, created through a collaboration between the Institut de Mathématiques de Marseille and the Centre de Physique Théorique.

- *Supporter of the educational association [EP83](#) which fights for the blossoming of the scholar environment and tries to answer to special educational needs of each children.*

LANGUAGES

- French: Native
- English: Fluent
- Spanish: Basic
- Chinese: Debutant